3	**(a)**	**(i)**	Explain what is meant by an *ideal gas*.	
			....	
			....	[2]
			An ideal gas is one that obeys the ideal gas equation $pV = nRT$ at all values of temperature (T), pressure (p) and volume (V), where n is the number of moles of gas and R is the molar gas constant.	**1** **1**
		(ii)	Use the kinetic theory of gases to explain why when the volume of an ideal gas decreases at constant temperature, the pressure of the gas increases.	
			....	
			....	
			....	
			....	
			....	
			....	
			....	[4]
			When temperature is constant, the root-mean-square speed of the gas atoms is constant. Therefore, **when volume decreases,** the distance travelled by the atoms between successive collisions with a wall of the container decreases, leading to a **higher frequency of collisions between the gas atoms and the wall of the container**. A **higher frequency of collisions leads to greater total <u>rate</u> of <u>change</u> of momentum of all the molecules hitting a wall** at any instant in time and **thus larger force on the container**, thus a higher pressure.	**1** **1** **1** **1**

(b) A fixed amount of an ideal gas undergoes a cycle of changes A→B→C→A as shown in Fig. 3.1.

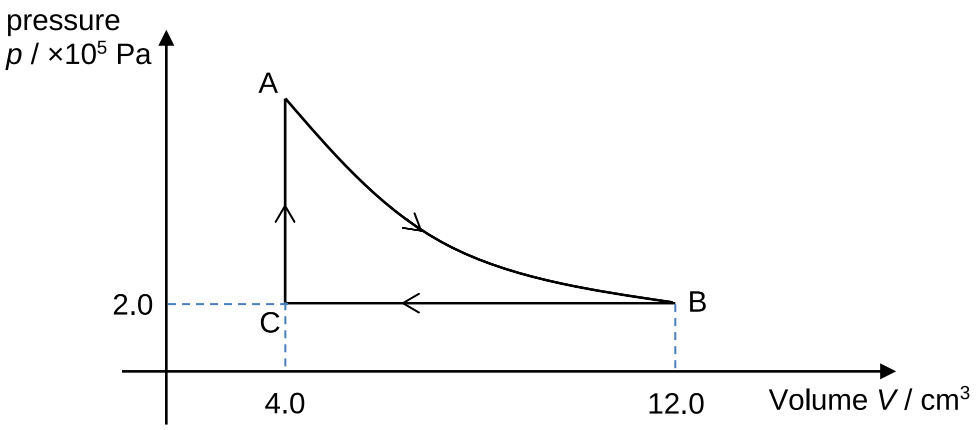

Fig. 3.1

(i) Determine the work done on the ideal gas during the process B→C.

work done on gas = ... J [2]

$W = -p\Delta V$

$W = -\left(2.0\times10^{5}\right)\left(4.0-12.0\right)\left(10^{-6}\right)$ **1**

$W = 1.6$ J **1**

(ii) Explain why there is a net thermal energy absorbed by the ideal gas when it undergoes a cycle of changes A→B→C→A.

....

....

....

....

....

....

.... [3]

Per cycle, **total** work done **by** gas is **greater** than **total** work done **on** the gas, thus there is a **net** work done **by** gas. **1**

Per cycle, there is **no net** change in the internal energy of the ideal gas. **1**

1

			Thus, by the First Law of Thermodynamics, $\Delta U = Q + W$, with a **net** work done **by** the gas, there needs to be a **net** thermal energy **supplied to** the gas so that there is **no net change in** the internal energy.	

4	The variation with potential difference V of current I in a resistor X is shown in Fig. 4.1.